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Maritime Vehicle with Removable Cap Having a Pick Point

Abstract

An autonomous maritime surface vehicle (AMSV) that includes a hull and a cap removably coupled to the hull (e.g., via a plurality of latches). The cap is movable between a first configuration, in which the cap is coupled to and sealingly encloses an entirety of an interior of the hull, and a second configuration in which the cap is at least partially decoupled from the hull, thereby permitting access to the interior of the hull. The AMSV also includes a single pick point coupled to a top, exterior surface of the cap, such that the AMSV can be easily and efficiently picked up and carried via the single pick point.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] The present application claims priority to each of the following: U.S. Provisional Patent Application No. 63/561,301, titled “Vision System for Maritime Vehicle,” and filed on Mar. 4, 2024; U.S. Provisional Patent Application No. 63/561,166, titled “Jet Pump Assembly for Maritime Vehicle” and filed on Mar. 4, 2024; U.S. Provisional Patent Application No. 63/561,063, titled “Power System for Maritime Vehicle” and filed on Mar. 4, 2024; U.S. Provisional Patent Application No. 63/561,282, titled “Systems and Approaches for Assembling a Maritime Vehicle” and filed on Mar. 4, 2024; U.S. Provisional Patent Application No. 63/561,181, titled “Micro-Keel Cooler for Maritime Vehicle” and filed on Mar. 4, 2024; and U.S. Provisional Patent Application No. 63/561,102, titled “Maritime Vehicle with Recessed Tow Points” and filed on Mar. 4, 2024. The contents of each of these disclosures is hereby incorporated by reference in their entirety.

FIELD OF THE DISCLOSURE
[0002] The present disclosure generally relates to maritime vehicles and more specifically to a maritime vehicle with a removable cap having a pick point.
BACKGROUND OF THE DISCLOSURE
[0003] Maritime vehicles, or vehicles designed for use on or in the water, are commonly used for transportation, recreation, defense, scientific research, and other purposes. Examples of maritime vehicles include boats, watercraft, submarines, and amphibious vehicles. Maritime vehicles can be manned (i.e., operated by an onboard human) or unmanned, and unmanned maritime vehicles can be remotely controlled or can be autonomous.
SUMMARY
[0004] An autonomous maritime surface vehicle (AMSV) that includes a hull and a cap removably coupled to the hull (e.g., via a plurality of latches). The cap is movable between a first configuration, in which the cap is coupled to and sealingly encloses an entirety of an interior of the hull, and a second configuration in which the cap is at least partially decoupled from the hull, thereby permitting access to the interior of the hull. The AMSV also includes one or more pick points coupled to a top, exterior surface of the cap, such that the AMSV can be easily and efficiently picked up and carried via the one or more pick points.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0005] The features of this invention which are believed to be novel are set forth with particularity in the appended claims. The invention may be best understood by reference to the following description taken in conjunction with the accompanying drawings, in which like reference numerals identify like elements in the several FIGS., in which:
[0006] FIG. **1** is a top perspective view of an example of a maritime vehicle constructed in accordance with the teachings of the present disclosure;
[0007] FIG. **2** is a side view of the maritime vehicle of FIG. **1**;
[0008] FIG. **3** is a front view of the maritime vehicle of FIG. **1**;
[0009] FIG. **4** is a rear view of the maritime vehicle of FIG. **1**;
[0010] FIG. **5** is a bottom perspective view of the maritime vehicle of FIG. **1**;

[0011] FIG. **6**A is similar to FIG. **1**, but with the cap and various components of the maritime vehicle removed for illustrative purposes;

[0012] FIG. **6**B is similar to FIG. **6**A, but with additional components of the maritime vehicle removed for illustrative purposes;

[0013] FIG. **6**C is a rear, perspective view of the maritime vehicle of FIG. **6**A;

[0014] FIG. **7**A is a top, perspective view of the cap of the maritime vehicle of FIGS. **1-5**;

[0015] FIG. **7**B is a bottom, perspective view of the cap of FIG. **7**A;

[0016] FIG. **7**C is a close-up view of a portion of the cap of FIG. **7**B;

[0017] FIG. **7**D is a cross-sectional view taken along line D-D in FIG. **7**A; and

[0018] FIG. **8** is a cross-sectional view taken along line **8-8** in FIG. **1**.

DETAILED DESCRIPTION

[0019] The present disclosure is directed to a maritime vehicle that is primarily intended for use for military purposes (e.g., for naval defense, patrolling waters and enforcing laws, reconnaissance, naval exploration, monitoring) but can also be used for other purposes if desired. The maritime vehicle is small(er), durable, and configured to quickly, efficiently, and stealthily traverse a body of water once dispatched (e.g., from other maritime vehicles, beachheads, or an airdrop). The maritime vehicle is modular, with components that can be flexibly altered, removed, or added as desired in accordance with the mission of the maritime vehicle. The maritime vehicle can collaborate with other similar maritime vehicles and/or military assets when necessary. The maritime vehicle is preferably unmanned and autonomous though need not be.

[0020] FIGS. **1-8** illustrate one example of a maritime vehicle **100** constructed in accordance with the teachings of the present disclosure. The maritime vehicle **100** is an unmanned vessel configured to autonomously traverse a body of water. In other words, the maritime vehicle **100** is an autonomous maritime surface vessel (or vehicle). The maritime vehicle **100** generally includes a hull **104** and a cap **108** that is coupled to the hull **104** to secure various components within the maritime vehicle **100**. The hull **104** is at least partially disposed in the body of water in which the maritime vehicle **100** is traversing. The hull **104** in this example is a mono-hull that has a front (or bow) **112**, a rear (or stern) **116**, two sides **120**, and a keel **124** coupled to another. The front **112**, the rear **116**, the sides **120**, and the keel **124** can be welded together or can be coupled to one another in a different manner. For example, the front **112**, the rear **116**, the sides **120**, and the keel **124** can be coupled together in the manner described in U.S. Provisional Application No. 63/561,282, titled “Systems and Approaches for Assembling a Maritime Vehicle” and filed Mar. 4, 2024, the contents of which are hereby incorporated by reference herein. The hull **104** is configured such that the hull provides a continuous planning surface that allows the maritime vehicle **100** to be highly maneuverable and to ride along the top of a body of water at high speeds, even in extreme weather conditions and difficult to navigate bodies of water. Meanwhile, the cap **108** is coupled to the hull **104** to cover and/or conceal the components of the maritime vehicle **100** disposed in and carried by the hull **104** as the maritime vehicle **100** traverses the body of water.

[0021] In this example, the cap **108** is removably coupled to the hull **104** so as to form a clamshell configuration that selectively covers and/or conceal the components of the maritime vehicle **100** disposed in and carried by an interior **126** of the hull **104**. More particularly, the cap **108** is reconfigurable, relative to the hull **104**, between a first configuration or position (shown in FIGS. **1-4** and **8**) and a second configuration or position (not explicitly illustrated). In the first configuration, the cap **108** is coupled to and sealingly encloses at least a portion of the interior **126** of the hull **104**. In this example, when the cap **108** is in the first configuration, the cap **108** sealingly encloses an entirety of the interior **126** of the hull **104**, such that the cap **108** entirely covers and conceals the components of the maritime vehicle **100** disposed in and carried by the hull **104**. At the same time, the cap **108** substantially prevents any water from the body of water and objects in the water (or the ambient environment more generally) from entering the interior **126** of the hull **104** via the cap **108** (or an interface between the hull **104** and the cap **108**). In other examples, however, the cap **108** may only sealingly enclose part of (and partially cover) the interior **126** when the cap **108** is in the first configuration. In the second configuration, the cap **108** is at least partially decoupled and spaced from the hull **104**, thereby permitting access to the interior **126** of the hull **104**. In some examples, the cap **108** will be completely decoupled, or removed, from the hull **104** in the second configuration (in which case the hull **104** will generally appear as shown in FIGS. **7**A-**7**C). In other examples, however, the cap **108** may be partially coupled to and partially decoupled, or separated, from the hull **104** in the second configuration, in which case the cap **108** may still partially cover but still permit access to the interior **126** of the hull **104**. For example, a first portion of the cap **108** may engage or be immediately adjacent to a first portion of the hull **104** while a second portion of the cap **108** is spaced from a second portion of the hull **104** (and the entire hull **104** more generally).

[0022] In this example, the hull **104** and the cap **108** each have a length that is equal to approximately 6 feet. In other examples, however, the length can vary. For example, the length can be equal to approximately 14 feet. The hull **104** is preferably entirely made of aluminum but can be partially or entirely be made of fiberglass and/or one or more other materials. In other examples, the maritime vehicle **100** can include two or more hulls (e.g., two parallel hulls) instead of the mono-hull.

[0023] As best illustrated in FIGS. **6**A-**6**C, the hull **104** also includes a perimeter **130** and a projection **132** that extends around at least a portion of the perimeter **130**. The perimeter **130** surrounds (and partially defines) the interior **126** and is generally defined by the front (or bow) **112**, the rear (or stern) **116**, and the two sides **120** of the hull **104**. In this example, the projection **132** extends outward (upward in the orientation shown in FIGS. **6**A-**6**C) from the front **112**, the rear **116**, and the two sides **120** around an entirety of the perimeter **130**. In other examples, however, the projection **132** may only extend around part of the perimeter **130** or the hull **104** may include two or more discrete projections **132** that each extend around a portion of the perimeter **130**. As illustrated in FIG. **6**B, the projection **132** is centrally located along the perimeter **130** such that the projection **132** defines a first, outer, recess **136** between the projection **132** and an exterior surface of the hull **104** and a second, inner recess **140** between the projection **132** and an interior surface of the hull **104**. It will be appreciated that each of the first and second recesses **136**, **140** is therefore configured to receive first and second portions of the cap **108** when the cap **108** is in the first configuration, as will be discussed in greater detail below.

[0024] As best illustrated by FIGS. **7**A-**7**C, the cap **108** has a shape and size that generally matches that shape and size of the hull **104**. More particularly, the cap **108** includes a perimeter **144** that has a shape and size that generally matches the shape and size of the perimeter **130** of the hull **104**. The cap **108** also includes a channel **148** that is defined by and between outer and inner walls **152**, **156** and matches the shape and size of the projection **132** of the hull **104**. As such, the channel **148** extends around at least a portion of the perimeter **144** of the cap **108**. In this example, the channel **148** extends around an entirety of the perimeter **144**. In other examples, however, the channel **148** may only extend around part of the perimeter **144** of the cap **108** or the cap **108** may include two or more discrete channels **148** that each extend around a portion of the perimeter **130**.

[0025] As best illustrated by FIGS. **7**A-**7**D and **8**, the cap **108** also includes a sealing element **160** disposed in the channel **148**. The sealing element **160** preferably takes the form of a bulb gasket made of rubber, as seen in FIGS. **7**D and **8**, though in other examples, the sealing element **160** can be a different type of seal (e.g., an O-ring, a metal gasket). In any event, the sealing element **152** also extends around at least a portion of the perimeter **144** of the cap **108**. Thus, in this example, the sealing element **160** extends around an entirety of the perimeter **144** of the cap **108**. In another example, e.g., when the channel **148** only extends around part of the perimeter **144**, the sealing element **160** may only extend around part of the perimeter **144**. In yet another example, e.g., when the cap **108** includes two or more discrete channels **148**, the cap **108** may include two or more sealing elements **160** disposed in the two or more discrete channels **148**, respectively.

[0026] It will be appreciated that when the cap **108** is in the first configuration (shown in FIGS. **1-4** and **8**), the perimeter **144** of the cap **108** is aligned with the perimeter **130** of the hull **104** such that the projection **132** of the hull **104** is partially disposed in the channel **148** of the cap **108**. In turn, the projection **132** of the hull **104** sealingly engages the sealing element **160** disposed in the channel **148** while the outer and inner walls **152**, **156** of the cap **108** occupy at least a portion of the outer and inner recesses **136**, **140**, respectively. Optionally, and in this example, the maritime vehicle **100** also includes an additional seal **164** that is coupled to the maritime vehicle **100** so as to surround an interface between a bottom surface of the cap **108** (in this case the walls **152**, **156**) and a top surface of the hull **104** (in this case the projection **132**) when the cap **108** is in the first configuration. In this example, the additional seal **164** is a Z-shaped rubber seal having one portion secured to the hull **104** and another portion secured to the cap **108**. In other examples, however, the additional seal **164** can have a different shape and/or can be made of a different material. In any case, these components help to substantially prevent any water from the body of water and objects in the water (or the ambient environment more generally) from entering the interior **126** of the hull **104** or an interior of the cap **108** via the cap **108** or the interface between the hull **104** and the cap **108**.

[0027] Conversely, it will be appreciated that when the cap **108** is in the second configuration, the projection **132** of the hull **104** is at least partially disposed outside of, or spaced from, the channel **148** of the cap **108**. In some examples, the entire projection **132** will be disposed outside of the entire channel **148** of the cap **108**. In other examples, part of the projection **132** will be disposed within part of the channel **148**, respectively, and a remainder of the projection **132** will be disposed outside of a remainder of the channel **148**. In any event, when the cap **108** is in the second configuration, the interior **126** of the hull **104** (and the interior of the cap **108**) are accessible. It will also be appreciated that in other examples, the cap **108** may include the projection **132** and the hull **104** may include the channel **148** and the sealing element **160**, and the cap **108** would still be reconfigurable between the first and second configurations in the manner described herein.

[0028] The cap **108** is generally removably coupled to the hull **104** via a hinge system that facilitates movement of the cap **108** between the first and second configurations described herein and yet helps to sealing enclose the interior **126** of the hull **104** when the cap **108** is in the first configuration. In this example, and as illustrated in FIGS. **1**, **3**, **4**, and **8**, the hinge system takes the form of a plurality of latches **200** disposed around at least a portion (if not the entirety) of a perimeter of the maritime vehicle **100**. In this example, the latches **200** are disposed on both sides **120** of the hull **104** and both sides of the cap **108** but not on or along the front and rear sides of the hull **104** or the cap **108**, and each of the latches **200** has two ends-a first end **202** and a second end **204** opposite the first end **202**—as well as an arm **206** that extends between and connects the first and second ends **202**, **204**. In this example, the first end **202** of each latch **200** is fixedly coupled (e.g., via one or more fasteners) to an exterior surface of one of the sides **120** of the hull **104**, whereas the second end **204** of each latch **200** is fixedly coupled (e.g., via one or more fasteners) to an exterior surface of one of the sides of the cap **104**. When the cap **108** is in the first configuration, and when the first and second ends **202**, **204** are fixedly coupled to the hull **104** and the cap **108**, respectively, the arm **206** of each latch **200** applies a tension force on both the first and second ends **202**, **204** so as to help maintain the sealed engagement between the hull **104** and the cap **108** discussed above. At the same time, the first end **202** and/or the second end **204** of one or more latches **200** can be removed from the hull **104** and/or the cap **108**, respectively, which allows the cap **108** to be moved from the first configuration to the second configuration. In some cases, some of the latches **200** can be removed from the hull **104** and/or the cap **108** so as to allow the cap **108** to be rotated relative to the hull **104** while the cap **108** is still secured to the hull **104** (via the remaining latches **200**). In other cases, each of the latches **200** can be removed from the hull **104** and/or the cap **108** so as to allow the cap **108** to be completely removed and separated from the hull **104**. In either case, the cap **108** is manipulated to allow access to the interior **126** of the hull **104**.

[0029] It will be appreciated that in other examples, the hinge system can facilitate the desired movement of the cap **108** relative to the hull **104** without being removed from the hull **104** or the cap **108**. For example, the latches **200** can facilitate the desired movement of the cap **108** without being removed from the hull **104** or the cap **108**. As another example, the latches **200** may only be positioned along one side **120** of the hull **104** and one side of the cap **108**, in which case the other side of the cap **108** is freely movable relative to the other side of the hull **104** via the latches **200**. It will also be appreciated that in other examples, the hinge system can be replaced by a locking system that serves to permanently couple the cap **108** to the hull **104** to permanently conceal or cover the components within the interior **126** of the hull **104**.
[0030] The maritime vehicle **100** further includes one or more pick points that allow the maritime vehicle **100** to be picked up and carried from one location (e.g., land) to another location (e.g., the body of water) without special tooling. In this example, the maritime vehicle **100** includes a single, load bearing pick point **210** that is coupled to a top, exterior surface **214** of the cap **108**. In this example, the single, load bearing pick point **210** is defined by a U-shaped bolt that is coupled to the top, exterior surface **214** of the cap **108**. Preferably, the single pick point **210** is located at or immediately adjacent to a center of gravity of the maritime vehicle **100** (when the cap **108** is in the first configuration). In turn, the maritime vehicle **100** can be safely and easily picked up via the single pick point **210** and transported as needed. In other examples, however, the maritime vehicle **100** may include multiple pick points coupled to the cap **108** (e.g., forward and rearward parts of the top, exterior surface **214**). In yet other examples, the maritime vehicle **100** may include a single pick point coupled to a portion of the hull **104** and/or may include one or more pick points coupled to the hull **104** and one or more pick points coupled to the cap **108**.
[0031] The maritime vehicle **100** also includes a plurality of bulkheads **220** arranged within the hull **104**. The bulkheads **220** divide the maritime vehicle **100** into a plurality of different compartments for receiving and retaining different components in the maritime vehicle **100**.
[0032] The maritime vehicle **100** also includes a sensor system that is generally configured to collect data about various components of the maritime vehicle **100** as well as data about the environment surrounding the maritime vehicle **100** (including data about objects in that environment). To this end, the sensor system generally includes a plurality of sensors disposed on an exterior or an interior of the maritime vehicle **100**. The sensors can include, for example, one or more pressure sensors (e.g., positioned to detect the pressure of the ambient air external to the maritime vehicle **100**, the pressure of the water in which the maritime vehicle **100** is disposed, the pressure within the maritime vehicle **100**), one or more temperature sensors (e.g., positioned to measure a temperature of a component of the maritime vehicle **100**, a temperature of ambient air external to the maritime vehicle **100**, a temperature of water in which the maritime vehicle **100** is disposed), one or more acoustic sensors (e.g., sonar sensors), one or more LIDAR sensors, one or more location sensors (e.g., GPS sensors, compass sensors), one or more motion sensors (e.g., accelerometers, gyroscopes), one or more infrared sensors, one or more water sensors (e.g., a float switch, a capacitive sensor, an ultrasonic sensor, an electrical water sensor, etc.) to determine when water is present and/or present to a given extent (e.g., at a certain volume or level), one or more humidity sensors, one or more power sensors (e.g., configured to detect charging or fueling levels), one or more lighting sensors (e.g., daylight sensors), one or more imaging sensors (e.g., CCD sensors, CMOS sensors), one or more magnetic sensors, or combinations thereof. Further details regarding the sensor system are described in U.S. Provisional Application Nos. 63/698,453, titled “Autonomous Maritime Surface Vehicles” and filed on Sep. 24, 2024, and 63/701,166, titled “Autonomous Maritime Surface Vehicles” and filed on Sep. 30, 2024, the contents of which are each hereby incorporated by reference in its entirety.
[0033] The maritime vehicle **100** also includes a power system that is generally configured to power the maritime vehicle **100** (and the components of the maritime vehicle **100**). The power system generally includes a thrust system and one or more power sources configured to power the

thrust system (and the other components within the maritime vehicle **100**). The thrust system is generally configured to propel the maritime vehicle **100** in/on/along the water. The thrust system can be a propeller-based thrust system or can be a jet pump-based thrust system such as the jet pump assembly described in U.S. Provisional Application No. 63/561,166, titled "Jet Pump Assembly for Maritime Vehicle" and filed Mar. 4, 2024, the contents of which are hereby incorporated by reference herein in its entirety. The one or more power sources can include, for example, one or more batteries, fuel (e.g., gasoline, diesel) stored in tanks carried by the maritime vehicle **100**, hydrogen stored in hydrogen tanks carried by the maritime vehicle **100**, solar panels (e.g., mounted to an exterior of the vehicle **100**), or other sources. The maritime vehicle **100** illustrated in FIGS. **1-8** includes four battery assemblies each including a rechargeable battery. The maritime vehicle **100** illustrated in FIGS. **1-8** also includes a retention assembly for the four battery assemblies, e.g., the retention assembly described in U.S. Provisional Application No. 63/561,063, titled "Power System for Maritime Vehicle" and filed Mar. 4, 2024, the contents of which are hereby incorporated by reference herein in its entirety. The maritime vehicle **100** generally also includes a cooling system configured to cool the thrust system and/or the one or more power sources, thereby preventing these components from overheating and leading to failure of the maritime vehicle **100**. For example, the maritime vehicle **100** can include the cooling system described in U.S. Provisional Application No. 63/561,181, titled "Micro-Keel Cooler for Maritime Vehicle" and filed Mar. 4, 2024, the contents of which are hereby incorporated by reference herein in its entirety.

[0034] In operation, the maritime vehicle **100** may be used to deploy and/or retrieve payloads such as, for example, persons, weapons (e.g., drones, missiles, mines, bombs), cargo (e.g., food), scientific instruments, or other equipment. Payloads can be deployed aerially (into the air), underwater, or on the surface of the water. Payloads can also be retrieved from the air, from underwater, or the surface of the water. Payloads to be deployed can be disposed in the hull **104**, attached to the exterior surface of the hull **104**, or attached to the exterior surface of the cap **108** prior to deployment. Likewise, retrieved payloads can be stored in the hull **104**, attached to and stored on the exterior surface of the hull **104**, or attached to and stored on the exterior surface of the cap **108**.

[0035] The maritime vehicle **100** can also include other systems to help with the operation of the maritime vehicle **100**, for example a ballast system, a navigation system, and a vision system. The ballast system is generally configured to stabilize the maritime vehicle **100** in the water, regardless of whether the maritime vehicle **100** is stationary or on the move. To this end, the maritime vehicle **100** may include one or more ballast tanks or chambers selectively filled with water or air to vary the buoyancy of the maritime vehicle **100**. Alternatively or additionally, the ballast system may include and utilize one or more inflatable devices to vary the buoyancy of the maritime vehicle **100**. The ballast system may also provide for the selective submerging and re-surfacing of the maritime vehicle **100** in a similar manner. The navigation system, which may for example be an inertial navigation system, utilizes the sensors of the sensor system to track the position and orientation of the maritime vehicle **100** and to guide the maritime vehicle **100** to its desired location in the body of water (or in a different body of water). In some examples, and as illustrated in FIGS. **1-3**, the vision system may be sealingly mounted to a front surface of the cap **108**. The vision system is generally configured to capture, process, and analyze images obtained by one or more cameras (e.g., a stereo camera and/or an infrared camera) and other data (e.g., data obtained by other sensors in the sensor system). The vision system can in turn identify or classify the environment surrounding the maritime vehicle **100** (including objects in that environment). Further details regarding the vision system are described in U.S. Provisional Application No. 63/561,301, titled "Vision System for Maritime Vehicle," and filed on Mar. 4, 2024, the contents of which are hereby incorporated by reference herein in its entirety.

[0036] The maritime vehicle **100** further includes a communications system that is generally configured to facilitate communication (i) between the maritime vehicle **100** and one or more central (remote) controllers, (ii) between the maritime vehicle **100** and and/or one or more other maritime vehicles **100** and/or other military assets (e.g., planes, ships), and (iii) between different components of the maritime vehicle **100**. The communications system generally includes one or more local (on-board) controllers and one or more communication modules (e.g., one or more antennae, one or more receivers, one or more transmitters, one or more radios, one or more ethernet switches) to effectuate wired or wireless communication between the maritime vehicle **100** and the central controller(s) or other maritime vehicles **100**. For example, the maritime vehicle **100** includes a plurality of antennae **228** disposed on the exterior of the cap **108** as well as a plurality of antennae **232** disposed in the hull **104**.

[0037] The one or more local controllers are generally configured to communicate data (e.g., operational instructions, data from the sensor system, data from other maritime vehicles **100** or military assets) and to perform automated operations of the maritime vehicle **100** based on that data. More particularly, the one or more local controllers are configured to control the maritime vehicle **100** based on that data. In some examples, the maritime vehicle **100** includes a plurality of different local controllers. For example, the maritime vehicle **100** can include one or more thrust controllers (for controlling the operation of the thrust system), one or more sensor controllers (for controlling the sensors in the sensor system), one or more payload controllers (for deploying or retrieving payloads), one or more navigation controllers (as part of the navigation system), and one or more ballast controllers (for controlling the ballast system). It will be appreciated that each of the one or more controllers may be implemented as hardware (e.g., processor, die, integrated device), software (e.g., non-transitory processor readable medium), and/or combinations thereof, in one or more devices (e.g., processor, chip, computer, tablet, mobile device).

[0038] While not explicitly described or illustrated herein, it will be appreciated that the maritime vehicle **100** includes several additional components. For example, the maritime vehicle **100** includes various sealing elements configured to provide seals between different components of the vehicle **100** (or between the vehicle **100** and the environment surrounding the vehicle **100**). As another example, the maritime vehicle **100** also includes various fasteners that help to couple the components of the maritime vehicle **100** together. As yet another example, the maritime vehicle **100** includes cabling that helps to communicatively couple components of the maritime vehicle **100** together. Indeed, the maritime vehicle **100** can include electrical cabling conveyed with the cap **108** through the bulkheads **220** or retained with hard wiring. As yet another example, the maritime vehicle **100** includes various electrical components that help to operate the maritime vehicle **100**, e.g., one or more relay boards, one or more DC-DC converters, one or more supervisor boards, and/or one or more brain boards.

[0039] Finally, although certain maritime vehicles have been described herein in accordance with the teachings of the present disclosure, the scope of coverage of this patent is not limited thereto. On the contrary, while the invention has been shown and described in connection with various preferred embodiments, it is apparent that certain changes and modifications, in addition to those mentioned above, may be made. This patent covers all embodiments of the teachings of the disclosure that fairly fall within the scope of permissible equivalents. Accordingly, it is the intention to protect all variations and modifications that may occur to one of ordinary skill in the art.

Claims

1. An autonomous maritime surface vehicle (AMSV), the AMSV comprising: a hull; a cap removably coupled to the hull, the cap being reconfigurable between a first configuration, in which the cap is coupled to and sealingly encloses at least a portion of an interior of the hull, and a second configuration in which the cap is at least partially decoupled from the hull, thereby permitting

access to the interior of the hull.
2. The AMSV of claim 1, wherein in the first configuration the cap sealingly encloses an entirety of the interior of the hull.
3. The AMSV of claim 1, further comprising one or more pick points coupled to an exterior surface of the cap.
4. The AMSV of claim 1, wherein the hull is a mono-hull.
5. The AMSV of claim 1, further comprising a hinge system that removably couples the cap to the hull.
6. The AMSV of claim 5, wherein the hinge system includes a plurality of latches disposed around at least a portion of a perimeter of the AMSV.
7. The AMSV of claim 1, further comprising a plurality of bulkheads arranged within the hull and configured to separate the interior of the hull into separate compartments.
8. The AMSV of claim 1, wherein when the cap is in the first configuration, a bottom surface of the cap engages a top surface of the hull, further comprising a seal surrounding an interface between the bottom surface of the cap and the top surface of the hull.
9. The AMSV of claim 1, wherein the cap includes a channel that extends along at least a portion of a perimeter of the cap, the channel sized to receive a top surface of the hull when the cap is in the first configuration.
10. An autonomous maritime surface vehicle (AMSV), the AMSV comprising: a hull; a cap removably coupled to the hull, the cap being movable between a first configuration, in which the cap is coupled to and sealingly encloses an entirety of an interior of the hull, and a second configuration in which the cap is at least partially decoupled from the hull, thereby permitting access to the interior of the hull; and a single pick point coupled to a top, exterior surface of the cap.
11. The AMSV of claim 10, wherein the hull is a mono-hull.
12. The AMSV of claim 10, further comprising a hinge system that removably couples the cap to the hull.
13. The AMSV of claim 12, wherein the hinge system includes a plurality of latches disposed around at least a portion of a perimeter of the AMSV.
14. The AMSV of claim 10, further comprising a plurality of bulkheads arranged within the hull and configured to separate the interior of the hull into separate compartments.
15. The AMSV of claim 10, wherein when the cap is in the first configuration, a bottom surface of the cap engages a top surface of the hull, further comprising a seal surrounding an interface between the bottom surface of the cap and the top surface of the hull.
16. The AMSV of claim 10, wherein the cap includes a channel that extends along at least a portion of a perimeter of the cap, the channel sized to receive a top surface of the hull when the cap is in the first configuration.
17. An autonomous maritime surface vehicle (AMSV), the AMSV comprising: a hull, the hull including a projection that extends around at least a portion of a perimeter of the hull; a cap removably coupled to the hull, the cap including a channel that extends along at least a portion of the perimeter of the hull; a sealing element disposed in the channel of the cap; wherein the cap is movable between a first configuration, in which the projection of the hull is disposed in the channel of the cap and sealingly engages the sealing element, thereby sealingly enclosing at least a portion of an interior of the hull, and a second configuration, in which the projection of the hull is at least partially spaced from the channel of the cap, thereby permitting access to the interior of the hull.
18. The AMSV of claim 17, wherein when the cap is in the first configuration, further comprising a seal surrounding an interface between the projection of the hull and channel of the cap.
19. The AMSV of claim 17, further comprising a hinge system that removably couples the cap to the hull, wherein the hinge system includes a plurality of latches disposed around at least a portion of a perimeter of the AMSV.

20. The AMSV of claim 17, further comprising a vision system sealingly mounted to a front surface of the cap, wherein the vision system comprises a stereo camera and an infrared camera.